THESIS REVIEWER'S REPORT

I. IDENTIFICATION DATA

Thesis title:	**Self-Supervised Segmentation of Synaptic Structures in Fluorescence Microscopy**
Author's name:	**Veronika Anokhina**
Type of thesis:	bachelor
Faculty/Institute:	Faculty of Electrical Engineering (FEE)
Department:	Department of Computer Science
Thesis reviewer:	Ing. Martin Čapek, Ph.D.
Reviewer's department:	Light Microscopy Core Facility, Institute of Molecular Genetics of the Czech Academy of Sciences

II. EVALUATION OF INDIVIDUAL CRITERIA

Assignment **extraordinarily challenging**

How demanding was the assigned project?

The assigned project was highly demanding both technically and scientifically. The work required the integration of several advanced research areas, including deep learning for biomedical image analysis, self-supervised learning, semantic segmentation, representation learning, clustering, statistical evaluation, and software engineering. The author not only implemented and compared state-of-the-art methods (SwinUNETR, SimMIM, VICReg, MoBY, Spotiflow, Leiden clustering, PERMANOVA), but also designed and validated a complete end-to-end pipeline operating without manual annotations. Beyond the machine learning aspects, the author developed a modular Python package, implemented a web-based user interface using Django, prepared Docker-based deployment for reproducibility, and integrated the entire workflow into a usable software system. This level of software engineering is uncommon in bachelor's projects and significantly increased the scope of the work. Overall, the project demanded a broad range of competencies spanning image analysis, machine learning, statistics, software development, and biological data interpretation. The complexity and breadth of the work correspond more closely to a strong master's-level project than to a standard bachelor's thesis.

Fulfilment of assignment **fulfilled**

How well does the thesis fulfil the assigned task? Have the primary goals been achieved? Which assigned tasks have been incompletely covered, and which parts of the thesis are overextended? Justify your answer.

Some assigned goals were only partially fulfilled. While the segmentation pipeline was successfully implemented, its performance could not be rigorously validated due to the absence of manually annotated ground truth. The thesis openly acknowledges this limitation and discusses it in detail. Therefore, the segmentation objective is implemented and demonstrated, but not fully validated. Conversely, the thesis exceeds the original assignment in several areas, including self-supervised representation learning, clustering analysis, and the development of a deployable web application with Docker support. These extensions significantly increase the scientific and practical value of the work. The thesis fulfils the assigned objectives to a very high degree.

Methodology **outstanding**

Comment on the correctness of the approach and/or the solution methods.

The approach and the chosen solution methods are correct, well justified, and consistent with current state-of-the-art practice in biomedical image analysis. The author demonstrates a good understanding of both the machine-learning and biological aspects of the problem. The selection of SwinUNETR as the backbone architecture, together with self-supervised pretraining using SimMIM and VICReg, is well motivated by recent literature. The use of pseudo-labels addresses the practical limitation of missing manual annotations and represents a reasonable strategy for this type of data. The methods are applied correctly, and the conclusions are supported by the presented results.

Technical level **A - excellent.**

Is the thesis technically sound? How well did the student employ expertise in the field of his/her field of study? Does the student explain clearly what he/she has done?

The thesis is technically sound and demonstrates a high level of competence in the fields of machine learning, image analysis, and software engineering. The student demonstrates a solid understanding of the relevant literature and appropriately motivates the selected methods. The work shows good technical judgment, particularly in adapting state-of-the-art approaches to the specific challenges of fluorescence microscopy data. The thesis is generally well structured and clearly written. The methodology, implementation details, experimental setup, and results are described in sufficient detail to allow reproduction of the work. The student clearly explains both the achieved results and the limitations of the proposed approach, including the absence of manually annotated ground truth for quantitative validation. Thus, the author demonstrated the ability to independently solve a complex interdisciplinary problem.

Formal and language level, scope of thesis **A - excellent.**

Are formalisms and notations used properly? Is the thesis organized in a logical way? Is the thesis sufficiently extensive? Is the thesis well-presented? Is the language clear and understandable? Is the English satisfactory?

The thesis is well organized, logically structured, and sufficiently extensive for a bachelor's thesis. Formalisms and notation are used correctly and consistently. The language is clear and easy to follow, and the English is of a very good standard. The thesis is professionally written and presented.

Selection of sources, citation correctness **A - excellent.**

Does the thesis make adequate reference to earlier work on the topic? Was the selection of sources adequate? Is the student's original work clearly distinguished from earlier work in the field? Do the bibliographic citations meet the standards?

The thesis makes adequate and extensive reference to relevant prior work. The selection of sources is appropriate and covers both the biological background and the current state of the art in machine learning, self-supervised learning, medical image segmentation, clustering, and software engineering. The student's original contributions are clearly distinguished from existing work, particularly in the design and implementation of the complete analysis pipeline, the experimental evaluation, and the integration of self-supervised learning, pseudo-label generation, clustering, and deployment into a unified framework. The bibliography is extensive, up-to-date, and contains a suitable mixture of foundational papers and recent publications. Bibliographic citations meet the expected academic standards.

Additional commentary and evaluation (optional)

Comment on the overall quality of the thesis, its novelty and its impact on the field, its strengths and weaknesses, the utility of the solution that is presented, the theoretical/formal level, the student's skillfulness, etc.

See below.

III. OVERALL EVALUATION, QUESTIONS FOR THE PRESENTATION AND DEFENSE OF THE THESIS, SUGGESTED GRADE

Summarize your opinion on the thesis and explain your final grading. Pose questions that should be answered during the presentation and defense of the student's work.

The thesis is of a high overall quality and significantly exceeds the scope typically expected from a bachelor's thesis. Its main strength lies not only in the implementation of a self-supervised segmentation pipeline, but especially in the demonstration that **self-supervised representations can capture biologically meaningful differences between treatment groups without the use of manual annotations**.

The clustering analysis is particularly noteworthy. By combining SSL pretraining, feature extraction, Leiden clustering, and PERMANOVA-based statistical evaluation, **the author demonstrates that the learned representations contain treatment-related biological information**. This part of the work provides stronger evidence of the usefulness of the learned features than the segmentation results themselves, which cannot be quantitatively validated due to the absence of manually annotated ground truth.

The thesis combines modern machine-learning methods with solid software engineering practices, resulting in a complete and reproducible analysis framework rather than a standalone proof-of-concept model. The implementation of a modular Python package, Docker deployment, and a web-based user interface further increase the practical utility of the work.

From a scientific perspective, the work successfully integrates recent advances in self-supervised learning, Transformer-based architectures, weak supervision, clustering, and statistical analysis. The theoretical level is appropriate, the methodology is well motivated, and the experiments are carefully designed and critically discussed.

The main weakness is the lack of manually annotated data, which prevents rigorous evaluation of segmentation accuracy. However, the author explicitly acknowledges this limitation and interprets the results appropriately.

Overall, the thesis demonstrates a high degree of technical skill, independence, and scientific maturity. Its originality lies primarily in **the integration of self-supervised representation learning and unsupervised biological phenotype discovery within a single end-to-end microscopy analysis pipeline**.

Questions:

1. The thesis describes the imaging and computational pipeline in detail, but the biological origin of neuronal cultures is not entirely clear. Were the neurons of murine, rat, or even human origin analyzed, and how might this affect the biological interpretation of the results?
2. What advantages did SwinUNETR provide over a conventional U-Net for this application, and do you believe the gains justify its increased complexity?
3. The clustering results appear to provide stronger evidence of biological relevance than the segmentation results. Why?
4. If you were given a small manually annotated subset of the data, what would be the first experiment you would perform and why?

The grade that I award for the thesis is **A - excellent. Outstanding.**

Date: **8.6.2026**

Signature: Čapek